

🌀 M003 Extension — The Loop Spiral (Intermediate)

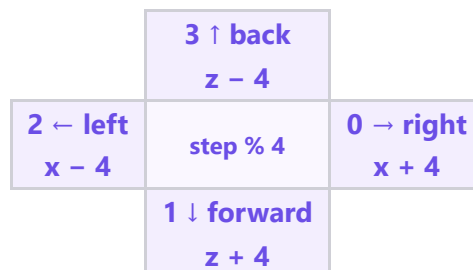
Topic: One loop that **turns** (x, y, z) · **Course:** 3D Coordinates · **Level:** Extension (Intermediate, after the Loop Staircase) · **Time:** about 40 minutes

The **staircase** took the **same** step every pass, so it grew in a **straight diagonal**. A **spiral** also **turns** every pass — same loop, but the direction **changes** — so the squares wrap into a tower. Type

SPIRAL in the chat and watch it **climb** and **turn**.

```
x = 8
y = 0
z = 8
for step in range(12):
    fill GOLD from (x, y, z) to (x + 2, y, z + 2)
    y = y + 1
    if step % 4 == 0:
        x = x + 4
    elif step % 4 == 1:
        z = z + 4
    elif step % 4 == 2:
        x = x - 4
    else:
        z = z - 4
```

 **Big idea:** `step % 4` counts **0, 1, 2, 3, 0, 1, 2, 3 ...** — four directions on repeat. That is the **turn**. `y = y + 1` lifts each square higher. **Turn + rise = a spiral.**



1 · Predict 🌟

Fill the **turn** for each step. `step % 4` repeats **0, 1, 2, 3**. Step 0 is done.

step	step % 4	turn (direction)
0	0	right (x + 4)
1		
2		
3		
4		
5		
6		
7		

`range(12)` builds how many squares? Circle one: 4 · 8 · 12

Which line makes the spiral go UP? Circle one: `y = y + 1` · `x = x + 4` · `step % 4`

Top-down, the 12 squares land on only 4 spots. Why only 4? (1 sentence)

2 · Spot the Bug 🐛

Bug A — no turn. A friend deleted the four `if` lines, so the direction never changes.

```
for step in range(12):  
    fill GOLD from (x, y, z) to (x + 2, y, z + 2)  
    y = y + 1  
    x = x + 4
```

What do you build instead of a spiral? Circle one: a straight diagonal · a flat square · nothing

Bug B — no rise. The `y = y + 1` line is gone, so y stays **0**.

What happens to the tower? Circle one: it stays flat on the floor · it grows taller · it turns upside down

Write the missing line and say where it goes:

3 · Show Your Work 📷 🎥

Load the world. Stand on your home spot. Type **SPIRAL** and watch each square **turn** to a new corner and **climb** one block.

👁️ **Top-down view (looking DOWN):** x → across, z ↓ deeper

	8	9	10	11	12	13	14
8							
9							
10							
11							
12							
13							
14							

Looking down, the squares sit on 4 corners — the loop turns right, forward, left, back, then repeats, climbing each time.

Modify challenge. Change **one** thing and **predict before you run** — for example `range(12)` → `range(20)` (taller), or the turn step `+ 4` → `+ 6` (wider). Write what will change, then build it.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what the loop does and how `step % 4` turns it.

2 · Your build. Point the camera at the spiral. Show it climbing and turning.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit 

Send this worksheet + your **one** video to teacher on KakaoTalk.